

Vishal Kumar Allam

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PROFESSIONAL SUMMARY

- Data Engineering professional with **6+ years of experience** designing, developing, deploying, and supporting scalable cloud-based data platforms and production pipelines using Python, SQL, PySpark, Apache Spark, Microsoft Azure, Databricks, and Snowflake.
- Strong hands-on experience building and optimizing **ETL/ELT, batch, and real-time data pipelines** with Azure Data Factory, Azure Databricks, Snowflake, Airflow, and distributed-processing frameworks for reporting, Advanced Analytics, and AI/ML use cases.
- Experienced implementing **data transformation, cleansing, validation, anomaly detection, reconciliation, metadata, lineage, governance, and quality controls** to ensure high levels of data integrity, consistency, transparency, and reliability.
- Deep expertise in **data modeling, Snowflake schema design, partitioning, indexing, clustering, caching, SQL optimization, Spark performance tuning, and workload analysis** for scalable and cost-efficient analytical workloads.
- Collaborated with **data scientists, analytics translators, business users, BI teams, product owners, and engineering stakeholders** to translate business needs into reusable data assets, APIs, feature-engineering datasets, and production data solutions.
- Strong production engineering background covering **Azure RBAC, security, compliance, Git, CI/CD, Terraform, ARM templates, logging, alerting, monitoring, troubleshooting, root-cause analysis, deployment automation, and production support**.
- Experienced owning solutions end to end across **design, development, deployment, monitoring, performance optimization, production support, enhancements, documentation, knowledge sharing, and continuous adoption of new data engineering technologies and architectures**.

EDUCATION

University of Missouri Kansas City

"Master of Science" in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

"Bachelor of Technology" in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **Programming & Data Processing:** Python, SQL, PySpark, Scala, Java, Bash, Apache Spark, Spark SQL, Pandas, Distributed Processing, Data Manipulation, Feature Engineering.
- **Azure & Cloud Data Platform:** Microsoft Azure, Azure Data Factory, Azure Databricks, Azure SQL, Azure Storage, Azure RBAC, Snowflake, Cloud Architecture, Cloud Storage, Compute Optimization.
- **Data Engineering:** ETL/ELT, Batch Pipelines, Real-Time Pipelines, Data Ingestion, Data Transformation, Data Cleansing, Data Modeling, Data Validation, Anomaly Detection, Data Reconciliation, Reusable Data Assets.
- **Performance & Governance:** Partitioning, Indexing, Clustering, Caching, Query Optimization, Spark Performance Tuning, Cost Optimization, Metadata, Data Lineage, Data Governance, Data Quality, Compliance.
- **Databases & Formats:** Snowflake, SQL Server, PostgreSQL, MongoDB, Cassandra, Relational/NoSQL Databases, JSON, Avro, Parquet, XML, Schema Design, Database Design.
- **DevOps & Production Engineering:** Git, Azure DevOps, CI/CD, Terraform, ARM Templates, Infrastructure as Code, Airflow, Docker, Deployment Automation, Logging, Monitoring, Alerting, Root-Cause Analysis, Production Support.
- **Analytics & Integration:** REST APIs, API Development, BI Enablement, Reporting, Advanced Analytics, AI/ML Data Pipelines, Model Deployment Support, Data Accessibility, Stakeholder Collaboration, Technical Documentation.

Test Automation Engineer

Oct 2024 – Present

PCMI LLC (Full Time)

Dallas, TX, USA

- Designed and delivered scalable **Azure and Snowflake data engineering solutions** using Python, SQL, PySpark, Azure Data Factory, Azure Databricks, and Palantir Foundry, supporting Advanced Analytics, operational reporting, digital products, and AI-driven insurance use cases.
- Built and optimized **ETL/ELT, batch, and near-real-time pipelines** to ingest data from APIs, relational databases, cloud storage, and enterprise applications, applying reusable orchestration patterns through ADF, Databricks, Spark, and Airflow concepts.
- Developed reusable **Python, PySpark, and SQL transformations** for cleansing, enrichment, normalization, reconciliation, feature engineering, and curated data preparation while maintaining high standards for completeness, accuracy, and consistency.
- Designed scalable **Snowflake and lakehouse data models** using partitioning, clustering, indexing strategies, caching, query tuning, and optimized Spark transformations to improve analytical performance and control cloud compute and storage costs.
- Implemented **data validation, anomaly detection, metadata, lineage, governance, security, and traceability controls**, working with business users and analytics teams to investigate data-quality issues and resolve root causes across pipelines.
- Automated infrastructure and production delivery using **Git, Azure DevOps, CI/CD, Terraform, ARM-template concepts, Azure RBAC, logging, monitoring, alerting, and deployment workflows**, improving security, reliability, and operational consistency.
- Owned solutions across **design, development, deployment, monitoring, troubleshooting, production support, enhancements, and knowledge sharing**, partnering with data scientists, analysts, product owners, and engineering teams to deliver reusable data assets and APIs.

Environments: Python, SQL, PySpark, Apache Spark, Azure Data Factory, Azure Databricks, Azure SQL, Snowflake, Palantir Foundry, ETL/ELT, Airflow, Data Modeling, Feature Engineering, Data Validation, Anomaly Detection, Metadata, Lineage, Azure RBAC, Terraform, ARM Templates, Git, Azure DevOps, CI/CD, Monitoring, Alerting.

Software Engineer

Dec 2023 – Sept 2024

Client: CVS

Dallas, TX, USA

- Designed and developed cloud-based **healthcare data pipelines and analytical services** using Python, SQL, PySpark, Spark, Azure/AWS services, and relational databases to support reporting, Advanced Analytics, and machine-learning initiatives.
- Built scalable **batch and event-driven ingestion workflows** that processed healthcare data from REST APIs, SQL databases, cloud storage, and enterprise applications, transforming raw information into reliable downstream analytical datasets.
- Developed Python, PySpark, and SQL transformations for **data cleansing, enrichment, validation, reconciliation, feature preparation, and aggregation**, improving the accessibility and consistency of healthcare data for analytical consumers.
- Implemented **data quality and anomaly-detection controls** for missing records, schema issues, duplicate data, reconciliation breaks, and inconsistent values while documenting metadata and data lineage for production workflows.
- Improved query and processing efficiency through **partitioning, indexing, caching, Spark tuning, SQL optimization, workload analysis, and scalable data-modeling techniques**, reducing processing latency across large datasets.
- Applied production engineering practices using **Git, Jenkins, CI/CD, Docker, cloud IAM/RBAC, monitoring, logging, alerting, deployment automation, and root-cause analysis** across development and production environments.
- Collaborated with **data scientists, analytics teams, architects, and business stakeholders** to gather requirements, deliver reusable data assets, support feature engineering and model consumption, improve BI accessibility, and resolve production issues.

Environments: Python, SQL, PySpark, Apache Spark, AWS, Azure Concepts, Azure Data Factory Concepts, Databricks Concepts, ETL/ELT, REST APIs, PostgreSQL, MySQL, Data Modeling, Data Validation, Anomaly Detection, Feature Engineering, Git, Jenkins, CI/CD, Docker, Monitoring, Alerting, Production Support.

Software Engineer

Sept 2021 – Dec 2022

Infor

Hyderabad, India

- Designed and developed enterprise **data engineering and analytics solutions** using Python, SQL, PySpark, Apache Spark, relational databases, and cloud-native technologies, supporting large-scale reporting and operational data workloads.
- Built reusable **ETL/ELT and orchestration pipelines** to extract data from APIs, databases, files, and enterprise applications and transform it into standardized datasets for analytics, reporting, and downstream applications.
- Implemented scalable processing using **PySpark, Spark SQL, Python, complex SQL, partitioning, caching, indexing, and optimized join strategies**, improving throughput and reducing processing bottlenecks across high-volume datasets.
- Designed relational and analytical **data models and schemas** across PostgreSQL, MySQL, Oracle, and SQL platforms, supporting efficient querying, historical reporting, data manipulation, and reusable consumption patterns.
- Implemented **data validation, cleansing, anomaly identification, reconciliation, metadata documentation, and data-quality controls** to improve accuracy, consistency, and traceability across integrated enterprise data flows.
- Applied engineering and DevOps practices using **Git, Jenkins, CI/CD, Docker, deployment automation, structured logging, monitoring, documentation, and production diagnostics** to improve operational reliability.
- Partnered with architects, analysts, product teams, and stakeholders to **translate business requirements into scalable solutions, optimize existing workloads, troubleshoot failures, support reporting needs, and contribute reusable data-processing components**.

Environments: Python, SQL, PySpark, Apache Spark, PostgreSQL, MySQL, Oracle, ETL/ELT, Data Modeling, Partitioning, Indexing, Caching, Data Validation, Reconciliation, REST APIs, Git, Jenkins, CI/CD, Docker, Logging, Monitoring, Production Support.

Associate Engineer

June 2019 – August 2021

Virtusa

Hyderabad, India

- Developed enterprise **data-processing and integration solutions** using Python, SQL, PostgreSQL, MySQL, REST APIs, and backend services, supporting operational reporting, analytics, and business-critical data workflows.
- Built reusable **ETL and batch-processing pipelines** to ingest information from relational databases, flat files, APIs, and enterprise systems and transform raw records into structured datasets for downstream consumers.
- Developed Python and SQL logic for **data cleansing, transformation, validation, reconciliation, enrichment, and automation**, reducing manual processing and improving the consistency of recurring data workflows.
- Optimized database workloads using **schema design, indexing, query tuning, joins, aggregations, caching concepts, and efficient data structures**, improving performance across transaction-heavy enterprise applications.
- Worked with structured and semi-structured formats including **JSON, XML, CSV, and relational records**, implementing integration logic for APIs, database services, and downstream reporting applications.
- Supported reliable delivery through **Git, CI/CD, Linux, Docker, deployment workflows, structured logging, monitoring, troubleshooting, documentation, and production incident resolution**.
- Collaborated with engineers, architects, analysts, and business stakeholders to **gather requirements, resolve data-quality issues, analyze production failures, improve workflow automation, and deliver maintainable data solutions**.

Environments: Python, SQL, PostgreSQL, MySQL, REST APIs, ETL, Batch Processing, Data Transformation, Data Validation, Data Reconciliation, JSON, XML, Data Modeling, Indexing, Query Optimization, Git, CI/CD, Docker, Linux, Logging, Monitoring, Production Support.

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.

- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.